Session2: Mixed-Effects Regression

aka multilevel or hierarchical regression

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Outline

1. Correlated data
2. Random intercept model
3. General mixed-effects model
4. Inference

Fixed-effects regression

Recall the **fixed-effects regression** model (session 1) $\mathrm{E}(y|\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{\beta}$ where $\boldsymbol{x}_i \in \mathbb{R}^p$ are the explanatory variables, and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^\top$, i.e.,

$$\boldsymbol{y} \quad = \quad \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \tag{1.1}$$

with

$$\boldsymbol{y} : n \times 1, \quad \boldsymbol{X} = \begin{pmatrix} \boldsymbol{x}_1^\top \\ \vdots \\ \boldsymbol{x}_n^\top \end{pmatrix} \sim n \times p, \quad \boldsymbol{\epsilon} : n \times 1.$$

Remember the assumptions for the error term (independency of the responses). there was is no structure in $\boldsymbol{\epsilon}$ and it represented the variations that could not be controlled (considered) in our studies.

Correlated data

If the independency is violated, the fixed-effect regression model (1.1) cannot be used any more. This may happen when we consider correlated responses:

- Measurements on clusters of subjects, e.g., a survey of all the students attended in LMO720 (cluster by the class).
- Repeated measurements on same subject, e.g., measurements on one patient that receives different treatments (it can be also seen as clustered by the patient!).
- Longitudinal data, e.g., measurements on one patient that receives a sequence of different treatments over a short or long period of time (Note that the crossover study is a longitudinal study which is a popular repeated-measures).
- Panel data, e.g., the measurements over different firms (sections) along a specific period of time (It is cross-sectional and longitudinal simultaneously).

Remedy for correlated responses

Recall the mixed-effect model is aka hierarchical model. If we assume that the data are being analyzed are drawn from a hierarchy of different populations or clustered whose differences relate to that hierarchy, then we deal with a random-effects model (effects has some levels but we only sample a random collection). In this case, we will treat the level of effects as samples from a random variable. Thus, we will consider a mixed-effects model as

$$\text{Mixed-effect model} = \text{Fixed-effects model} + \text{Random-effects model} \tag{1.2}$$

Fixed-eff.	gender(male/female), condition(wet-dry), dosage(low/high)
Random-eff.	family, individuals with repeated measures, classrooms

Linear Mixed-effects Model

Adopting (1.2), the linear mixed-effects model (LMM) in the nature of mean regression is given by

$$\mathrm{E}(Y_{ij}|\boldsymbol{x}_{ij}, \boldsymbol{z}_{ij}) = \underbrace{\boldsymbol{x}_{ij}^\top \boldsymbol{\beta}}_{\text{fixed-effects component}} + \underbrace{\boldsymbol{z}_{ij}^\top \boldsymbol{b}}_{\text{random-effects component}}, \tag{1.3}$$

with

- Y_{ij} = response of j-th member of cluster i, $i = 1, \cdots, n, j = 1 \cdots, n_i$
- $\boldsymbol{x}_{ij}$ = covariate vector of j-th member of cluster i for fixed effects, $\boldsymbol{\beta} \in \mathbb{R}^p$
- $\boldsymbol{z}_{ij}$ = covariate vector of j-th member of cluster i for random effects, $\boldsymbol{b} \in \mathbb{R}^q$

Matrix notation of LMM-1

Including the measurement error component, the model (1.3) can be represented in matrix notation as

$$\boldsymbol{Y}_i = \boldsymbol{X}_i \boldsymbol{\beta} + \boldsymbol{Z}_i \boldsymbol{b}_i + \boldsymbol{\epsilon}_i, \tag{1.4}$$

where $\boldsymbol{Y}_i = (Y_{i1}, \cdots, Y_{in_i})^\top : n_i \times 1$, $\boldsymbol{X}_i = (\boldsymbol{x}_{i1}^\top, \cdots, \boldsymbol{x}_{in_i}^\top)^\top : n_i \times p$, $\boldsymbol{Z}_i = (\boldsymbol{z}_{i1}^\top, \cdots, \boldsymbol{z}_{in_i}^\top)^\top : n_i \times q$, $\boldsymbol{\epsilon}_i = (\epsilon_{i1}, \cdots, \epsilon_{in_i})^\top : n_i \times 1$.

Here $\boldsymbol{X}_i$ and $\boldsymbol{Z}_i$ are the fixed-effects and random-effects design matrices, respectively, for cluster i. Remember we have n clusters.

Matrix notation of LMM-2

The final framework for the LMM is

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{Z}\boldsymbol{b} + \boldsymbol{\epsilon}, \tag{1.5}$$

where ($N = \sum_{j=1}^{n} n_j$)

$$\boldsymbol{Y} = \begin{pmatrix} \boldsymbol{Y}_1 \\ \vdots \\ \boldsymbol{Y}_n \end{pmatrix} : N \times 1, \ \boldsymbol{X} = \begin{pmatrix} \boldsymbol{X}_1 \\ \vdots \\ \boldsymbol{X}_n \end{pmatrix} \sim N \times p, \ \boldsymbol{b} = \begin{pmatrix} \boldsymbol{b}_1 \\ \vdots \\ \boldsymbol{b}_n \end{pmatrix} : nq \times 1$$

$$\boldsymbol{Z} = \begin{pmatrix} \boldsymbol{Z}_1 & & 0 \\ & \ddots & \\ 0 & & \boldsymbol{Z}_n \end{pmatrix} : N \times nq, \quad \boldsymbol{\epsilon} = \begin{pmatrix} \boldsymbol{\epsilon}_1 \\ \vdots \\ \boldsymbol{\epsilon}_n \end{pmatrix} : N \times 1. \tag{1.6}$$

Random Intercept Model

The random intercept model assumes that each cluster/group affects the responses through only the intercept term of the cluster. The normal model has form

$$Y_{ij} = \alpha + \boldsymbol{x}_{ij}\boldsymbol{\beta} + a_i + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, \sigma_\epsilon^2) \perp a_i \sim N(0, \sigma_a^2) \tag{2.1}$$

where Y_{ij} is the response for the j-th individual in the i-th cluster/group, α is the fixed intercept (constant), a_i is the random intercept associated with cluster i, and x_{ij} is the vector of predictors associated with $\boldsymbol{\beta} = (\beta_1, \cdots, \beta_p)^\top$; $j = 1, \cdots, n_i$, $i = 1, \cdots, n$.
Why the random intercept model is a mixed-effects model?

Characteristics

Description: Since $Y_{ij}|a_i$ is an ordinary linear model with intercept $\alpha + a_i$, we use the terminology "random intercept". This model models the effect of clustering using cluster-specific intercepts.

Mean: $\mathrm{E}(Y_{ij}|\boldsymbol{x}_{ij}) = \alpha + \boldsymbol{x}_{ij}\boldsymbol{\beta}$.

Covariance: $\sigma_{jk} = \mathrm{Cov}(Y_{ij}, Y_{ik}) = \begin{cases} \mathrm{Var}(Y_{ij}) = \sigma_a^2 + \sigma_\epsilon^2 & j = k \\ \sigma_a^2 & j \neq k \end{cases}$

Correlation In one cluster, the correlation between the responses is called "intraclass correlation" and is equal to $\rho = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_\epsilon^2}$.

Estimation

For estimation of $\boldsymbol{\beta}$ under the likelihood paradigm, we maximize the likelihood function $\Pr(\boldsymbol{Y}|\boldsymbol{X}, \alpha, \boldsymbol{\beta}, \sigma^2_{\boldsymbol{a}}, \sigma^2_{\boldsymbol{\epsilon}})$.

Strategy=REML, aka residual or reduced maximum likelihood:

1. We first transform the dataset so that the likelihood function of the transformed dataset depends only on $\boldsymbol{\Sigma} = (\sigma_{jk})$, but not on $\boldsymbol{\beta}$ and then estimate the covariance matrix.
2. Once $\boldsymbol{\Sigma}$ is estimated, we estimate $\boldsymbol{\beta}$.

(I will only touch at a later time!)

Lab3

We generate a sample from the random intercept model

$$Y_{ij} = \alpha + \beta X_{ij} + a_i + \epsilon_{ij} \tag{2.2}$$

We use the package "nlme": Linear and Nonlinear Mixed Effects Models
There are several datasets in the package—Let's look at the pdf file from the source
CODE:

```
rm(list = ls())
alpha = -2
beta = 2
n = c(10, 15, 12, 15, 12, 15)
npeople = length(n)
...
```

Lab3-1

We use the package "lme4": Linear Mixed-Effects Models using 'Eigen' and S4

Distribution of $\boldsymbol{Y}_i$

Recall the model (1.4)

$$\boldsymbol{Y}_i = \boldsymbol{X}_i\boldsymbol{\beta} + \boldsymbol{Z}_i\boldsymbol{b}_i + \boldsymbol{\epsilon}_i,$$

with $\boldsymbol{Y}_i = (Y_{i1}, \cdots, Y_{in_i})^\top : n_i \times 1$, $\boldsymbol{X}_i = (\boldsymbol{x}_{i1}^\top, \cdots, \boldsymbol{x}_{in_i}^\top)^\top : n_i \times p$, $\boldsymbol{Z}_i = (\boldsymbol{z}_{i1}^\top, \cdots, \boldsymbol{z}_{in_i}^\top)^\top : n_i \times q$, $\boldsymbol{\epsilon}_i = (\epsilon_{i1}, \cdots, \epsilon_{in_i})^\top : n_i \times 1$.
We allocate the specific covariances

$$\mathrm{Cov}(\boldsymbol{b}_i) = \boldsymbol{D}_i : q \times q$$
$$\mathrm{Cov}(\boldsymbol{\epsilon}_i) = \boldsymbol{\Sigma}_i : n_i \times n_i$$

Then, the covariance of the model is given by
$\boldsymbol{V}_i = \mathrm{Cov}(\boldsymbol{Y}_i) = \boldsymbol{Z}_i\boldsymbol{D}_i\boldsymbol{Z}_i^\top + \boldsymbol{\Sigma}_i$.

Specifically, we assume

$$\boldsymbol{\epsilon}_i \overset{iid}{\sim} N_{n_i}(\mathbf{0}, \boldsymbol{\Sigma}_i) \perp \boldsymbol{b}_i \overset{iid}{\sim} N_q(\mathbf{0}, \boldsymbol{D}_i) \tag{3.1}$$

Consequently, we have

$$\boldsymbol{Y}_i \overset{iid}{\sim} N_{n_i}(\boldsymbol{X}_i\boldsymbol{\beta}, \boldsymbol{V}_i), \quad \boldsymbol{V}_i = \boldsymbol{Z}_i\boldsymbol{D}_i\boldsymbol{Z}_i^\top + \boldsymbol{\Sigma}_i \tag{3.2}$$

Distribution of $\boldsymbol{Y}$

Recall the model (1.5)

$$\boldsymbol{Y} = \boldsymbol{X}\beta + \boldsymbol{Z}\boldsymbol{b} + \boldsymbol{\epsilon} \tag{3.3}$$

Under the assumptions (3.1), we have

$$\begin{pmatrix} \boldsymbol{b} \\ \boldsymbol{\epsilon} \end{pmatrix} \sim N_{nq+N}\left(\begin{pmatrix} \mathbf{0}_{nq\times 1} \\ \mathbf{0}_{N\times 1} \end{pmatrix}, \begin{pmatrix} \boldsymbol{D}_{nq\times nq} & \mathbf{0}_{nq\times N} \\ \mathbf{0}_{N\times nq} & \boldsymbol{\Sigma}_{N\times N} \end{pmatrix} \right) \tag{3.4}$$

Remember $N = \sum_{j=1}^{n} n_j$.
Here

$$\boldsymbol{D} = \begin{pmatrix} \boldsymbol{D}_1 & & 0 \\ & \ddots & \\ 0 & & \boldsymbol{D}_n \end{pmatrix} \text{ and } \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_1 & & 0 \\ & \ddots & \\ 0 & & \boldsymbol{\Sigma}_n \end{pmatrix} \tag{3.5}$$

Distribution of $\boldsymbol{Y}$ cont.

Finally, we have the following fixed-effects (say) representation for the mixed-effects model (3.3)

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}^*, \tag{3.6}$$

where

$$\boldsymbol{\epsilon}^* = \boldsymbol{Z}\boldsymbol{b} + \boldsymbol{\epsilon} = \begin{pmatrix} \boldsymbol{Z}_{N\times nq} & \boldsymbol{I}_{N\times N} \end{pmatrix} \begin{pmatrix} \boldsymbol{b}_{nq\times 1} \\ \boldsymbol{\epsilon}_{N\times 1} \end{pmatrix} \tag{3.7}$$

with

$$\boldsymbol{\epsilon}^* \sim N_N(\boldsymbol{0}, \boldsymbol{V}), \tag{3.8}$$

$$\begin{aligned} \boldsymbol{V} &= \begin{pmatrix} \boldsymbol{Z} & \boldsymbol{I}_N \end{pmatrix} \begin{pmatrix} \boldsymbol{D} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{\Sigma} \end{pmatrix} \begin{pmatrix} \boldsymbol{Z}^\top \\ \boldsymbol{I}_N \end{pmatrix} \\ &= \begin{pmatrix} \boldsymbol{Z}\boldsymbol{D} & \boldsymbol{\Sigma} \end{pmatrix} \begin{pmatrix} \boldsymbol{Z}^\top \\ \boldsymbol{I}_N \end{pmatrix} \\ &= \boldsymbol{Z}\boldsymbol{D}\boldsymbol{Z}^\top + \boldsymbol{\Sigma} \end{aligned} \tag{3.9}$$

Lab4

We analyze the "sleepstudy" dataset with a LMM. This dataset is available in the R-package "lme4".
These data are from the study described in Belenky et al. (2003)[1], for the sleep-deprived group and for the first 10 days of the study, up to the recovery period.

[1] Belenky, G., Wesensten, N.J., Thorne, D.R., Thomas, M.L., Sing, H.C., Redmond, D.P., Russo, M.B. and Balkin, T.J. (2003). Patterns of performance degradation and restoration during sleep restriction and subsequent recovery: a sleep dose-response study, Journal of Sleep Research, 12, 1?12.

Lab4 cont.

Description

The average reaction time per day for subjects in a sleep deprivation study. On day 0 the subjects had their normal amount of sleep. Starting that night they were restricted to 3 hours of sleep per night. The observations represent the average reaction time on a series of tests given each day to each subject.

Format

A data frame with 180 observations on the following 3 variables:

`Reaction` Average reaction time (ms)

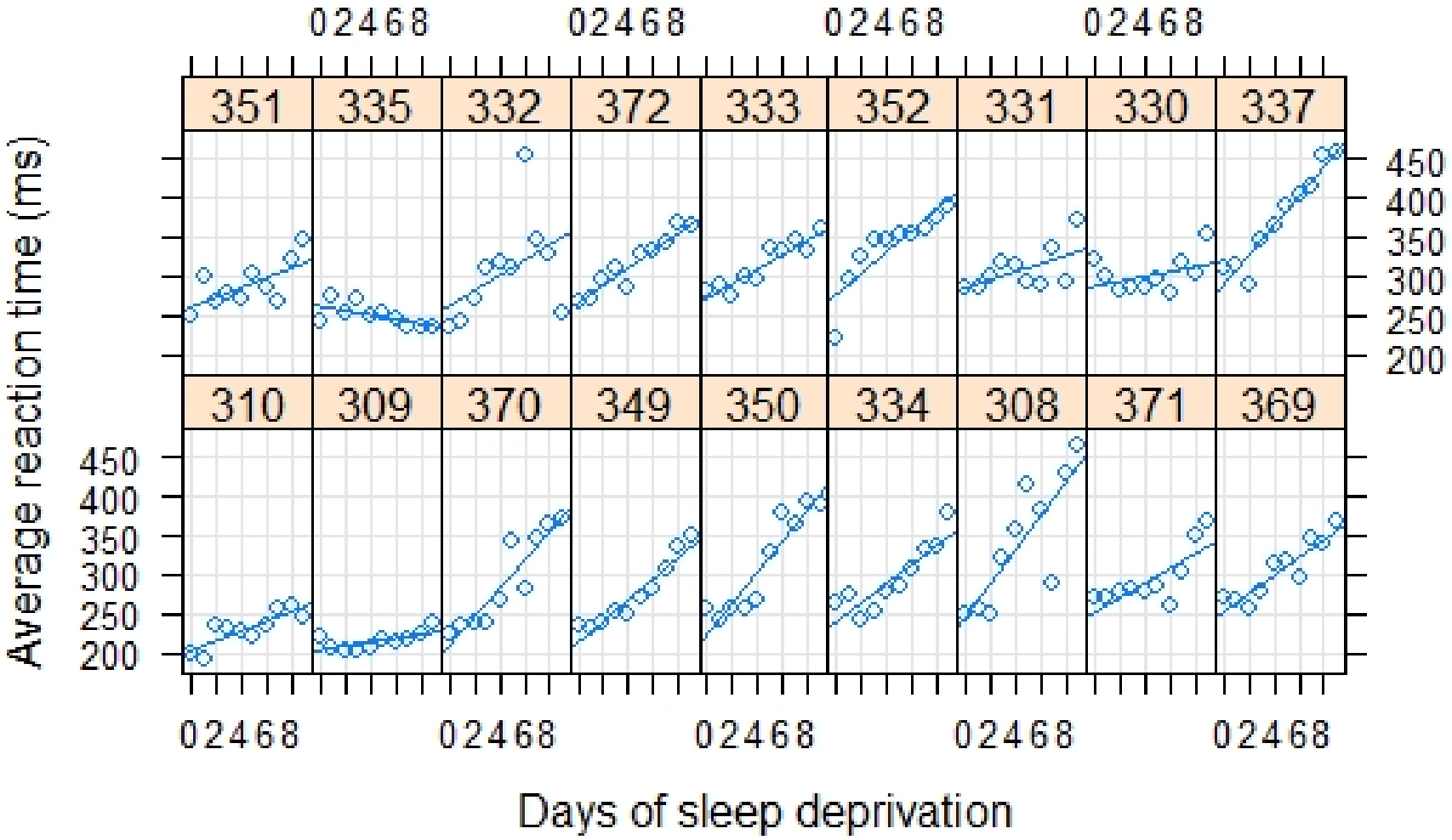
`Days` Number of days of sleep deprivation

`Subject` Subject number on which the observation was made

Lab4 cont.

```
> library(lme4)
> str(sleepstudy)
'data.frame':   180 obs. of  3 variables:
 $ Reaction: num  250 259 251 321 357 ...
 $ Days    : num  0 1 2 3 4 5 6 7 8 9 ...
 $ Subject : Factor w/ 18 levels "308","309","310",..: 1 1 1 1 1 1 1 1 1
> dim(sleepstudy)
[1] 180   3
> head(sleepstudy)
  Reaction Days Subject
1 249.5600    0     308
2 258.7047    1     308
3 250.8006    2     308
4 321.4398    3     308
5 356.8519    4     308
6 414.6901    5     308
> require(lattice)
> xyplot(Reaction ~ Days | Subject, sleepstudy, type = c("g","p","r"),
+        index = function(x,y) coef(lm(y ~ x))[1],
+        xlab = "Days of sleep deprivation",
+        ylab = "Average reaction time (ms)", aspect = "xy")
```

Lab4 cont.


351
335
332
372
333
352
331
330
337
310
309
370
349
350
334
308
371
369
Average reaction time (ms)
Days of sleep deprivation
0 2 4 6 8
200
250
300
350
400
450

Lab4 cont.

First we fit the following LMM

$$\text{lmer}(\text{Reaction} \sim \text{Days} + (\text{Days}|\text{Subject}), \text{sleepstudy}) \tag{3.10}$$

In this model, (Days|Subject) is the random effect term and Subject refers to the cluster/class/group index i. The output is:

```
Linear mixed model fit by REML ['lmerMod']
Formula: Reaction ~ Days + (Days | Subject)
   Data: sleepstudy
REML criterion at convergence: 1743.628
Random effects:
 Groups   Name        Std.Dev. Corr
 Subject  (Intercept) 24.741
          Days         5.922   0.07
 Residual             25.592
Number of obs: 180, groups:  Subject, 18
Fixed Effects:
(Intercept)         Days
     251.41        10.47
```

Lab4 cont.

More details are available under the summary command of the output as

```
> summary(fm1)
Linear mixed model fit by REML ['lmerMod']
Formula: Reaction ~ Days + (Days | Subject)
   Data: sleepstudy

REML criterion at convergence: 1743.6

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.9536 -0.4634  0.0231  0.4634  5.1793

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 Subject  (Intercept) 612.10   24.741
          Days         35.07    5.922   0.07
 Residual             654.94   25.592
Number of obs: 180, groups:  Subject, 18

Fixed effects:
            Estimate Std. Error t value
(Intercept)  251.405      6.825  36.838
Days          10.467      1.546   6.771

Correlation of Fixed Effects:
     (Intr)
Days -0.138
```

Lab4 cont.

Here, we analyze the output. First of all, we assumed normality!
For the fixed-effects part of the model we have

$$\begin{aligned} \hat{\beta}_0 &= \text{Intercept} = 251.405 \\ \hat{\beta}_1 &= \text{Days} = 10.467 \end{aligned} \tag{3.11}$$

Estimates of the variances (this is only for the random-effects part)

$$\begin{aligned} \hat{\sigma}^2_{b0} &= \text{random Intercept} = 612.10 \\ \hat{\sigma}^2_{b1} &= \text{random Days} = 35.07 \\ \hat{\rho} &= 0.07 \end{aligned} \tag{3.12}$$

Thus we assumed the following (random intercept and random slope) model

$$Y_{ij} = \beta_0 + \beta_1 \text{Day}_{ij} + b_{i0} + b_{i1}\text{Day}_{ij} + \epsilon_{ij} \tag{3.13}$$

$$\boldsymbol{b} = \begin{pmatrix} b_{i0} \\ b_{i1} \end{pmatrix} \overset{iid}{\sim} N_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2_{b0} & \rho\sigma_{b0}\sigma_{b1} \\ \rho\sigma_{b0}\sigma_{b1} & \sigma^2_{b1} \end{pmatrix}\right) \tag{3.14}$$

Lab4 cont.

Finally, we interpret the output.
Correlation between the Subject's intercept and slope (related to the random components) is estimated to be 0.07. It appears that a subject's response to sleep deprivation is not related much at all to their inherent reaction ability.

If we fit the random intercept (only) model and compare the two models we will see the results will support both the random intercept and slope model; (see Lab4-1 for details).

Estimation of parameters of the LMM

For only the estimation of $\boldsymbol{\beta}$, we can use the fixed-effects representation of the LMM. Recall the model (3.6)

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \underbrace{\boldsymbol{Z}\boldsymbol{b} + \boldsymbol{\epsilon}}_{\boldsymbol{\epsilon}^*}, \quad \boldsymbol{\epsilon}^* \sim N_N(0, \boldsymbol{V}), \quad \boldsymbol{V} = \begin{pmatrix} \boldsymbol{V}_1 & & 0 \\ & \ddots & \\ 0 & & \boldsymbol{V}_n \end{pmatrix}.$$

Note that $\boldsymbol{V}$ is a block-diagonal matrix with blocks $\boldsymbol{V}_i$ on the main diagonal and zeros elsewhere.

From the least square theory, the OLS estimate of $\boldsymbol{\beta}$ is given by

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{V}^{-1} \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{V}^{-1} \boldsymbol{Y}. \tag{4.1}$$

It can be shown that (4.1) is the MLE of $\boldsymbol{\beta}$ (see Assignment Q3).

Estimation of the random-effects $\boldsymbol{b}$

In order to estimator $\boldsymbol{b}$, we follow the steps bellow

Derive the joint distribution $(\boldsymbol{Y}^\top, \boldsymbol{b}^\top)^\top$

↓

Obtain $\mathrm{E}(\boldsymbol{b}|\boldsymbol{Y})$

↓

The estimate of $\boldsymbol{b}$ is $\mathrm{E}(\boldsymbol{b}|\boldsymbol{Y})$ when we replace $\hat{\boldsymbol{\beta}}$

From the facts that $\boldsymbol{Y} \sim N_N(\boldsymbol{X}\beta, \boldsymbol{V})$ and $\boldsymbol{b} \sim N_{nq}(0, \boldsymbol{D})$ we have

$$\begin{pmatrix} \boldsymbol{Y} \\ \boldsymbol{b} \end{pmatrix} \sim N_{N+nq}\left(\begin{pmatrix} \boldsymbol{X}\beta \\ \boldsymbol{0} \end{pmatrix}, \begin{pmatrix} \mathrm{Var}(\boldsymbol{Y}) & \mathrm{Cov}(\boldsymbol{Y}, \boldsymbol{b}) \\ \mathrm{Cov}(\boldsymbol{Y}, \boldsymbol{b})^\top & \mathrm{Var}(\boldsymbol{b}) \end{pmatrix} \right) \tag{4.2}$$

where

$$\begin{aligned} \mathrm{Cov}(\boldsymbol{Y}, \boldsymbol{b}) &= \boldsymbol{\Sigma}_{\boldsymbol{Yb}} \\ &= \mathrm{Cov}(\boldsymbol{X}\beta + \boldsymbol{Zb} + \boldsymbol{\epsilon}, \boldsymbol{b}) \\ &= \mathrm{Cov}(\boldsymbol{X}\beta, \boldsymbol{b}) + \mathrm{Cov}(\boldsymbol{Zb}, \boldsymbol{b}) + \mathrm{Cov}(\boldsymbol{\epsilon}, \boldsymbol{b}) \\ &= \boldsymbol{0} + \boldsymbol{Z}\,\mathrm{Var}(\boldsymbol{b}) + \boldsymbol{0} = \boldsymbol{ZD} \end{aligned} \tag{4.3}$$

Using equations (4.2) & (4.3) we can obtain the conditional distribution as

$$\boldsymbol{b}|\boldsymbol{Y} \sim N_{nq}\left(\boldsymbol{\mu}_{\boldsymbol{b}|\boldsymbol{Y}}, \boldsymbol{\Sigma}_{\boldsymbol{b}|\boldsymbol{Y}}\right), \tag{4.4}$$

where

$$\begin{aligned} \boldsymbol{\mu}_{\boldsymbol{b}|\boldsymbol{Y}} &= \boldsymbol{\mu}_{\boldsymbol{b}} + \boldsymbol{\Sigma}_{\boldsymbol{b}\boldsymbol{Y}}\boldsymbol{\Sigma}_{\boldsymbol{Y}}^{-1}(\boldsymbol{Y} - \boldsymbol{\mu}_{\boldsymbol{Y}}) \\ &= \boldsymbol{0} + \boldsymbol{D}\boldsymbol{Z}^{\top}\boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}) \qquad (4.5) \\ \boldsymbol{\Sigma}_{\boldsymbol{b}|\boldsymbol{Y}} &= \boldsymbol{\Sigma}_{\boldsymbol{b}} - \boldsymbol{\Sigma}_{\boldsymbol{b}\boldsymbol{Y}}\boldsymbol{\Sigma}_{\boldsymbol{Y}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{Y}\boldsymbol{b}} \\ &= \boldsymbol{D} - \boldsymbol{D}\boldsymbol{Z}^{\top}\boldsymbol{V}^{-1}\boldsymbol{Z}\boldsymbol{D} \qquad (4.6) \end{aligned}$$

Thus,

$$\begin{aligned} \mathrm{E}(\boldsymbol{b}|\boldsymbol{Y}) &= \boldsymbol{D}\boldsymbol{Z}^{\top}\boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}) \qquad (4.7) \\ \mathrm{Var}(\boldsymbol{b}|\boldsymbol{Y}) &= \boldsymbol{D} - \boldsymbol{D}\boldsymbol{Z}^{\top}\boldsymbol{V}^{-1}\boldsymbol{Z}\boldsymbol{D} \qquad (4.8) \end{aligned}$$

Using (4.7), we have

$$\hat{\boldsymbol{b}} = \boldsymbol{D}\boldsymbol{Z}^\top \boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}) \tag{4.9}$$

This is called the empirical best linear unbiased estimator (Why?)

Further notes on estimation

We did not simultaneously estimate $\boldsymbol{\beta}$ and $\boldsymbol{b}$. Using the hierarchical representation of the LMM (see Assignment Q2), we can obtain the likelihood as

$$\Pr(\boldsymbol{Y}, \boldsymbol{b} | \boldsymbol{X}, \boldsymbol{Z}) = f(\boldsymbol{Y}, \boldsymbol{b}) = f(\boldsymbol{Y} | \boldsymbol{b}) f(\boldsymbol{b}) \tag{4.10}$$

and differentiate w.r.t. the parameters to get system of normal equations; (we do not discuss further).

Recall the model (3.6) with the assumption (3.8). It implies the marginal model (equation (3.2))

$$\boldsymbol{Y}_i \overset{iid}{\sim} N_{n_i}(\boldsymbol{X}_i\boldsymbol{\beta}, \boldsymbol{V}_i), \quad \boldsymbol{V}_i = \boldsymbol{Z}_i\boldsymbol{D}_i\boldsymbol{Z}_i^\top + \boldsymbol{\Sigma}_i.$$

Let $\boldsymbol{\alpha}$ denote the vector of all variance and covariance parameters (usually called variance components) found in $\boldsymbol{V}_i$, that is, $\boldsymbol{\alpha}$ consists of the $q(q+1)/2$ different elements in D and of all parameters in $\boldsymbol{\Sigma}_i$. Finally, let $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, \boldsymbol{\alpha}^\top)^\top$ be the s-dimensional vector of all parameters in the marginal model for $\boldsymbol{Y}_i$, and let $\Theta = \Theta_{\boldsymbol{\beta}} \times \Theta_{\boldsymbol{\alpha}}$ denote the parameter space for $\boldsymbol{\theta}$, with $\Theta_{\boldsymbol{\beta}}$ and $\Theta_{\boldsymbol{\alpha}}$ the parameter spaces for the fixed-effects and for the variance components respectively. Note that $\Theta_{\boldsymbol{\beta}} = \mathbb{R}^p$, and $\Theta_{\boldsymbol{\alpha}}$ equals the set of values for $\boldsymbol{\alpha}$ such that $\boldsymbol{D}$ and all $\boldsymbol{\Sigma}_i$ are positive (semi-)definite.

The likelihood is given by

$$\prod_{i=1}^{n} \left\{ (2\pi)^{-\frac{n_i}{2}} |\boldsymbol{V}_i(\boldsymbol{\alpha})|^{-\frac{1}{2}} \right.$$
$$\left. \times \exp\left(-\frac{1}{2}(\boldsymbol{Y}_i - \boldsymbol{X}_i\boldsymbol{\beta})^\top \boldsymbol{V}_i^{-1}(\boldsymbol{\alpha})(\boldsymbol{Y}_i - \boldsymbol{X}_i\boldsymbol{\beta}) \right) \right\} \tag{4.11}$$

According to our main reference (Verbeke and Molenberghs, 2001), we assume $\boldsymbol{\alpha}$ is known and obtain the estimate of $\boldsymbol{\beta}$ as (as in (4.1))

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha}) = \left(\sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i^{-1}(\boldsymbol{\alpha}) \boldsymbol{X}_i \right)^{-1} \sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i^{-1}(\boldsymbol{\alpha}) \boldsymbol{Y}_i. \tag{4.12}$$

If we know the estimate of $\boldsymbol{\alpha}$, $\hat{\boldsymbol{\alpha}}$ say, then $\widehat{\boldsymbol{V}_i(\boldsymbol{\alpha})} = \boldsymbol{V}_i(\hat{\boldsymbol{\alpha}})$. Then, $\widehat{\boldsymbol{\beta}(\boldsymbol{\alpha})} = \hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\alpha}})$. One way of estimating $\boldsymbol{\alpha}$ is the RMLE.

Restricted maximum likelihood estimation (RMLE)

The RML estimator for the variance components $\boldsymbol{\alpha}$ is obtained from maximizing the likelihood function of a set of error contrasts $\boldsymbol{U} = \boldsymbol{A}^\top \boldsymbol{Y}$ where $\boldsymbol{A}$ is any $N \times (N-p)$ full-rank matrix with columns orthogonal to the columns of the $\boldsymbol{X}$ matrix. Using $\boldsymbol{Y} \sim N_N(\boldsymbol{X}\boldsymbol{\beta}, \boldsymbol{V})$,

$$\boldsymbol{U} \sim N_{N-p}(\boldsymbol{0}, \boldsymbol{A}^\top \boldsymbol{V}(\boldsymbol{\alpha})\boldsymbol{A}) \tag{4.13}$$

since $\boldsymbol{A}^\top \boldsymbol{X} = \boldsymbol{0}$. Refer to the main reference for more info.

Suggested study

I suggest to read a tutorial available at
"https://web.stanford.edu/class/psych252/section/Mixed_models_tutorial.html"
adopted from Winter, B. (2013). Linear models and linear mixed effects models in R with linguistic applications. arXiv:1308.5499.

Assignments

Q1 : Get the "pig.weights" data from the package "SemiPar". Fit a random intercept model and analyze the output.

Q2 : Using (3.4) derive the distribution of $\boldsymbol{Y}$ in (3.3) and show that linear mixed model (3.3) can be represented as the two level hierarchical model

$$\boldsymbol{Y}|\boldsymbol{b} \sim N_N(\boldsymbol{X}\beta + \boldsymbol{Z}\boldsymbol{b}, \Sigma), \quad \boldsymbol{b} \sim N_{nq}(\boldsymbol{0}, \boldsymbol{D}).$$

Q3 : Show that (4.1) is the maximum likelihood estimate of $\boldsymbol{\beta}$ in the model (3.6), under the normality assumption.

Assignments

Q4 : Consider the following example from the package "lme4"
Call the package "mlmRev": Examples from Multilevel Modelling Software Review
library(mlmRev)
Fit the following models and analyze the outputs in view of the differences.

$$\text{lmer}(\text{normexam} \sim 1 + (1|\text{school}), \text{data=Exam})$$

$$\text{lmer}(\text{normexam} \sim \text{standLRT} + (1|\text{school}), \text{data=Exam})$$

$$\text{lmer}(\text{normexam} \sim \text{standLRT} + (\text{standLRT}|\text{school}), \text{data=Exam})$$

Thank you for your attention.